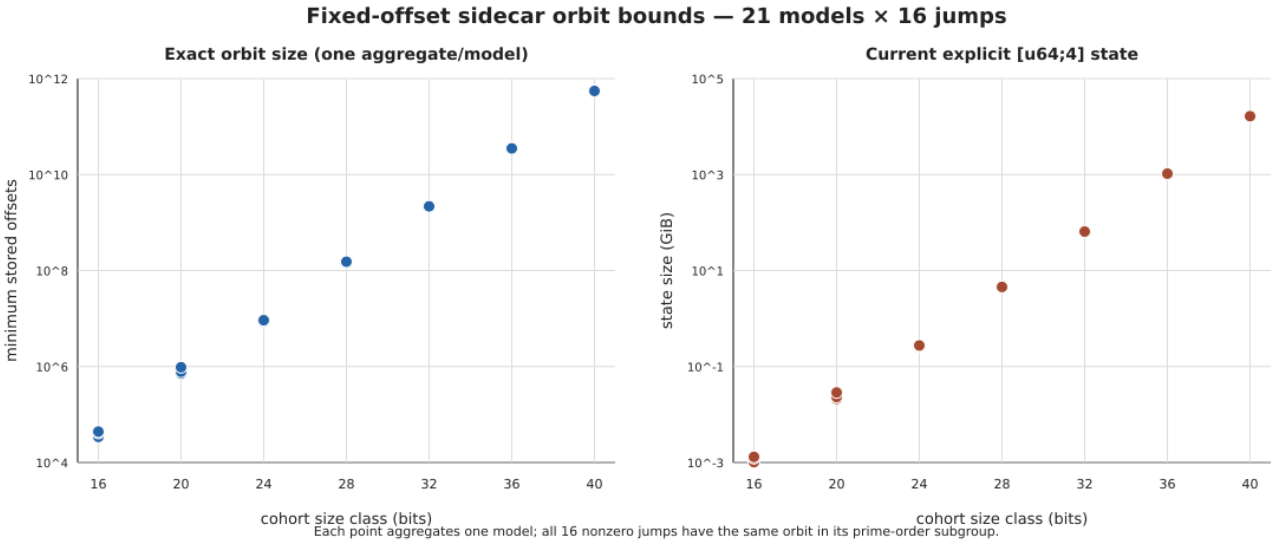
Fixed-offset Kummer sidecar orbit bounds

Study KUMMER-FIXED-OFFSET-ORBIT-20261024

24 October 2026

This proof-and-derivation run uses 21 frozen prime-order groups and 16 canonical nonzero jumps per group. It executes no target and performs no scalar recovery.

Recorded quantity	Result
Models / jumps / bound rows	21 / 16 / 336
Standard price rows	12,096
Exact full-order orbit rows	336 / 336
Minimum logical offsets	34,147–552,130,504,327
Current explicit state (bytes)	1,092,704–17,668,176,138,464
Regular cached-xADD bound (M-equivalent)	819,528–13,251,132,103,848
Native validator failures	0



Statement and scope. If a nonempty state explicitly stores $K(T+s)$ and a direct one-xADD update after fixed jump J requires the predecessor offset, closure under $s \mapsto s-J$ contains the whole translation orbit. Lean proves this conditional closure theorem. The native validator checks that every frozen subgroup order is prime and every jump scalar is nonzero and coprime to it, so all 336 orbit sizes equal r.

Storage uses the current Rust `[u64;4]` representation. Update cost uses 24 field multiplications and 36 field additions for one regular cached xADD per entry. These are representation- and recurrence-specific bounds, not a full collector price or a rho comparison. Compressed states, project-pseudomultiply-recover chains, label-free recurrences, variable-offset sidecars, and different walks remain outside the theorem.

Artifacts. The packet contains the frozen JSON input, 336-row bounds, 12,096-row price grid, independent validation receipt, graph, theorem source, proof-gate log, manifest, and SHA-256 list. The manifest pins the complete input digest.